T Ritvik Rohan Reddy

es24btech11035@iith.ac.in | ritvik.tabishala@gmail.com | +91 8861940418 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.82
Sri Chaitanya Techno School	Class XII (CBSE)	2024	95.60%
Sri Chaitanya Techno School	Class X (CBSE)	2022	90.60%

PROJECTS

System Design Playground | *FastAPI, ChromaDB, Cytoscape.js, SQLite, OpenRouter LLM*

- Engineered an AI-powered distributed systems platform supporting **7+ infrastructure components** and **20+ REST APIs**, enabling interactive graph-based architecture design, visualization, and persistent storage.
- Implemented a Retrieval-Augmented Generation (RAG) pipeline over **10+ distributed systems knowledge documents** to generate context-aware LLM-powered architecture reviews and optimization recommendations.
- Developed an automated analysis engine capable of identifying **50+ architecture anti-patterns**, including single points of failure, missing load balancers, absent cache layers, messaging gaps, and scalability bottlenecks.
- Architected a modular backend with layered services, graph-based domain modeling, and reusable abstractions to support architecture management, validation, and AI-assisted analysis.
- Bought together an interactive graph visualization interface with real-time component and connection management, seamlessly integrating architecture editing with backend analysis services.
- Applied production-style software engineering practices including clean separation of concerns, extensible modules, and maintainable APIs, enabling future expansion with authentication, collaboration, and cloud deployment.

CoinVision | PyTorch, FastAPI, ResNet18, Grad-CAM, SQLAlchemy, SQLite, HTML/CSS/JavaScript

- Engineered an end-to-end computer vision platform classifying **315+ coin categories** using a fine-tuned **ResNet18 Transfer Learning** model trained on **10,248 images**, achieving **70.88% validation accuracy** through staged fine-tuning, stratified validation, checkpointing, and data augmentation.
- Built a modular FastAPI backend exposing RESTful APIs for real-time inference, Top-3 confidence predictions, Grad-CAM generation, and prediction history using SQLAlchemy and SQLite.
- Implemented **Grad-CAM** from scratch using PyTorch forward and backward hooks to generate class activation heatmaps, providing interpretable visual explanations for model predictions.
- Designed an optimized inference pipeline with reusable preprocessing, confidence estimation, label decoding, and hardware-aware execution across CPU, CUDA, and Apple MPS devices.
- Developed an interactive frontend using HTML, CSS, and JavaScript for image upload, real-time prediction, Grad-CAM visualization, and prediction history, delivering a complete end-to-end machine learning application.

Additional Projects

- **Whispr** | Flask, SQLAlchemy, SQLite - Developed a full-stack messaging application featuring secure user authentication, private one-to-one messaging, profile management, and persistent message storage.
- **Movie Recommendation System** | Streamlit, scikit-learn, Pandas - Built a content-based movie recommendation engine using cosine similarity on the TMDB dataset, delivering real-time Top-5 movie recommendations.
- **Repository Classifier** | Flask, scikit-learn - Developed an NLP-powered application to automatically classify GitHub repositories into software domains using machine learning.

POSITIONS OF RESPONSIBILITY

Web Coordinator | EBSB Club, IIT Hyderabad

- Coordinated the development and maintenance of the club website, collaborating with design and content teams to deliver new features.
- Developed responsive frontend components using React, Next.js, and Tailwind CSS.
- Managed project collaboration through GitHub, coordinating feature development and code integration.
- Contributed to improving the website's usability, responsiveness, and overall user experience.

SKILLS

Programming Languages: C, C++, Python, SQL

Web Technologies: HTML, CSS, JavaScript

Frameworks & Libraries: FastAPI, Flask, SQLAlchemy, PyTorch, scikit-learn, Pandas, NumPy, OpenCV

Databases: PostgreSQL, SQLite, ChromaDB

Developer Tools: Git, GitHub, VS Code, Linux

Core Computer Science: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

RELEVANT COURSES

Computer Science: Data Structures and Applications, Algorithms, Theory of Computation, Discrete Mathematics, Operating Systems, Database Management Systems

Mathematics: Probability Random Variables, Statistics,Elementary Linear Algebra, Differential Equations, Numerical Methods, Complex Variables, Calculus

EXTRACURRICULAR

Sports: Cricket, Badminton, Football, Volleyball

A part of National Service Scheme (NSS)